Import Data Transform Map

(Spread Sheet)

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DEVELOPMENT PHASE

1. Introduction:

- The Development Phase is an important stage of the project in which the actual project components are created and configured according to the project requirements.
- In this project, "Import Data Using Transform Maps", the development process focuses on creating the required Service Now tables, fields, Import Set table, Transform Map, field mappings, data transformation, Coalesce configuration, reports, and dashboard.
- The main purpose of development is to create a complete employee data management process in Service Now. External employee data is prepared in an Excel spreadsheet and then imported into Service Now. The imported data is first stored in an Import Set table. A Transform Map is then used to transfer the data from the Import Set table to the Employee Test table.

The development process can be represented as:

Excel Spreadsheet → Import Set → Transform Map → Field Mapping → Employee Test Table → Reports → Dashboard

- The project is developed step by step so that employee information can be imported and managed efficiently.

2. Preparation of Employee Data:

- The first step in the development process is preparing the employee data.
- A new Google Spreadsheet is created and sample employee information is entered into the spreadsheet. The spreadsheet contains the required employee details that will later be imported into Service Now.

The main employee fields include:

- Employee ID
- Employee Name
- Email
- Department
- Location

- After entering the sample data, the spreadsheet is downloaded in Excel (.xlsx) format.
- This Excel file acts as the external source of employee data for the project.
- The prepared spreadsheet is important because it provides the data that will be loaded into Service Now during the import process.

3. Development of Custom Employee Table:

- After preparing the source data, the next step is to create a custom table in Service Now.

The table is created through:

- Tables → Create New

The following details are provide

- Label: Employee Test
- Name: u_employee_test

- After creating and saving the table, the form is opened using the Show Form option.
- The Form Layout option is then used to create the required fields for storing employee information.

The first field created is:

- Employee ID – String

- Additional fields are also created for storing employee information.

The required fields are:

- Employee ID
- Employee Name
- Email
- Department
- Location

- All these fields are created using the String type.
- After creating the fields, the form is saved and the newly created fields are verified on the Employee Test form.
- This custom table becomes the target table for the Transform Map

4. Development of Import Set Table:

- The next stage is creating the Import Set table
- The Import Set table acts as a temporary staging area for the data imported from the Excel spreadsheet.

The navigation used is:

- Application Navigator → Load Data
- The Load Data option is opened and the import table is created.

The following details are provided:

- Label: Employee Import
- Name: u_employee_import

- The name is automatically populated by the system.
- After providing the required information, the form is submitted.
- The Employee Import table is used as the source table for the Transform Map.
- The project documentation describes the spreadsheet data as being loaded into an Import Set table before it is transformed into the target table.

5. Development of Transform Map:

- The Transform Map is one of the main components of this project.
- It is used to transfer data from the Import Set table to the Employee Test table.
- After creating the Import Set table, the Create Transform Map option is selected.

The Transform Map is configured with the following details:

- Name: Sample Spreadsheet Import
- Source Table: Employee Import
- Target Table: Employee Test

- The source table is automatically populated as the Employee Import table.
- The Employee Test table is selected as the target table.
- The Transform Map establishes the relationship between the source data and the target data.
- After providing the required details, the Auto Map Matching Fields option is used.
- The Mapping Assist option is then used to map the source fields with the corresponding target fields.
- After completing the mapping, the Transform Map is saved.

6. Development of Field Mapping:

- Field mapping is an important part of the development process.
- The fields from the Employee Import table need to be connected to the corresponding fields in the Employee Test table.

The mapping is performed as follows:

Source Field	Target Field
Employee ID	Employee ID
Employee Name	Employee Name
Email	Email
Department	Department
Location	Location

- This mapping ensures that the employee information is transferred to the correct fields in the target table.
- For example, the Employee ID from the source spreadsheet is transferred to the Employee ID field in the Employee Test table.
- Similarly, employee name, email, department, and location are transferred to their respective target fields.
- Correct field mapping is necessary for maintaining the accuracy of the imported employee information.

7. Development of Data Transformation:

- After completing the Transform Map and field mapping, the data transformation process is developed.
- The Transform option is selected to execute the transformation.
- The transformation transfers employee information from the Employee Import table to the Employee Test table.
- During this process, the system processes the imported records according to the field mappings configured in the Transform Map.
- After clicking the Transform button, the system displays the transformation result.

The expected status is:

State: Completed

Status: Success

After successful transformation, the employee records become available in the Employee Test table.

This forms the main working part of the project.

8. Development of Coalesce Function:

- Another important feature developed in this project is Coalesce.
- Coalesce is used to identify existing employee records during repeated imports.
- In real-world situations, the same employee data may be imported multiple times. Without proper identification, duplicate records could be created.
- To avoid this problem, a field is configured as Coalesce in the Transform Map.
- In this project, the Employee ID is used for identifying the employee record.

The development process includes:

- Opening the created Transform Map.
- Opening the Field Maps section.
- Selecting the required field.
- Changing the Coalesce value from False to True.
- Saving the Transform Map.

- When an Employee ID already exists, the system can identify the existing record and update it instead of creating another record.
- When a new Employee ID is imported, a new employee record can be inserted.
- The project document specifically uses Coalesce to prevent duplicate employee records and update existing records with new values.

9. Development for New and Updated Employee Data:

- The project is also developed to handle both new and existing employee information.
- Additional employee data is prepared in Excel format.
- Some records contain existing Employee IDs with modified information such as name and email. New Employee IDs are also added.
- The new Excel data is loaded into the existing Employee Import table.

The import process uses:

- Existing Import Set table
- Excel file
- Sheet number
- Header row
- Existing Transform Map

- After submitting the data, the system processes the records
- New Employee IDs are inserted as new records, while existing Employee IDs are updated with the changed information.
- The documented project example processes four rows, where two new IDs are inserted and two existing records are updated.
- This makes the project useful for repeated employee data imports.

10. Development of Reports:

After completing the data import and transformation process, reports are developed to display employee information clearly.

Three reports are created using the Employee Test table.

10.1 Employees by Department

The first report is called:

- Employees by Department

The configuration includes:

- Source Type: Table
- Table: Employee Test
- Chart Type: Pie Chart
- Group By: Department
- Aggregation: Count

- This report displays the number of employees belonging to different departments.
- The report is executed and saved after configuration.

10.2 Employees by Location

The second report is:

- Employees by Location

The configuration includes:

- Table: Employee Test
- Type: Bar Chart
- Group By: Location
- Aggregation: Count

- This report provides employee information based on their locations.
- The report is executed and saved after completing the configuration.

10.3 Employee List Report

The third report is:

- Employee List Report
- The table selected is Employee Test.

The report displays the following columns

- Employee ID
- Employee Name
- Email
- Department
- Location

- The list report provides a simple view of employee records.
- After selecting the required columns, the report is executed and saved.

11. Development of Dashboard:

- After creating the reports, a dashboard is developed to display the reports in a centralized location.

The dashboard is created through:

- All Menus → Dashboards

A new dashboard is created with the name:

- Employee Analytics Dashboards
- The dashboard provides a single location for viewing the employee reports.

The three reports developed earlier are added to the dashboard:

- Employees by Department
- Employees by Location
- Employee List Report

- The reports are added using the Add to Dashboards option.
- The project documentation confirms that the three reports are added successfully to the created dashboard.

12. Integration of Reports and Dashboard:

- The reports and dashboard are integrated to provide a centralized view of employee information.
- The Employees by Department report provides department-wise employee information.
- The Employees by Location report provides location-wise employee information.
- The Employee List Report provides detailed employee records.
- These reports are combined into the Employee Analytics Dashboard.

The dashboard can therefore provide information such as:

- ❖ Total employee records
- ❖ Employee distribution by department
- ❖ Employee distribution by location
- ❖ Employee details
- ❖ Newly added employees
- ❖ Updated employee records
- ❖ Import-related information

- The project documentation states that the reports and dashboard provide meaningful insights and visibility into employee data and import activities.

13. Overall Development Flow:

The complete development process can be summarized as follows:



- This complete development process creates an automated employee data import and management system in Service Now.

14. Development Outcome:

- The development phase successfully creates all the major components required for the project.
- The Employee Test table provides the required structure for employee information.
- The Employee Import table acts as the staging area for external data.
- The Transform Map transfers data from the source table to the target table.
- Field mapping ensures that the employee information is placed in the correct fields.
- Coalesce helps identify existing employees and prevents unnecessary duplicate records.
- Reports provide different views of employee information.
- The Employee Analytics Dashboard combines the reports into a centralized view.

As a result, the project provides an organized solution for importing, updating, and monitoring employee information in Service Now.

15. Conclusion:

- The Development Phase successfully implements the main requirements of the Import Data Using Transform Maps project.
- The project begins with preparing employee information in an Excel spreadsheet. A custom Employee Test table is then created in Service Now with the required employee fields. An Employee Import table is developed to receive the external spreadsheet data.
- The Transform Map is developed to connect the source and target tables. Field mapping is configured to ensure that employee information is transferred to the correct fields. The transformation process then creates the required employee records in the target table.
- The project is further enhanced by developing Coalesce functionality. This allows the system to identify existing employee records and update them instead of creating duplicate records. The project also includes reports for department-wise data, location-wise data, and employee lists.
- Finally, an Employee Analytics Dashboard is developed to combine the reports into a single centralized view.
- Therefore, the development phase establishes the complete working structure of the project and provides an efficient method for importing, updating, and viewing employee data in Service Now.